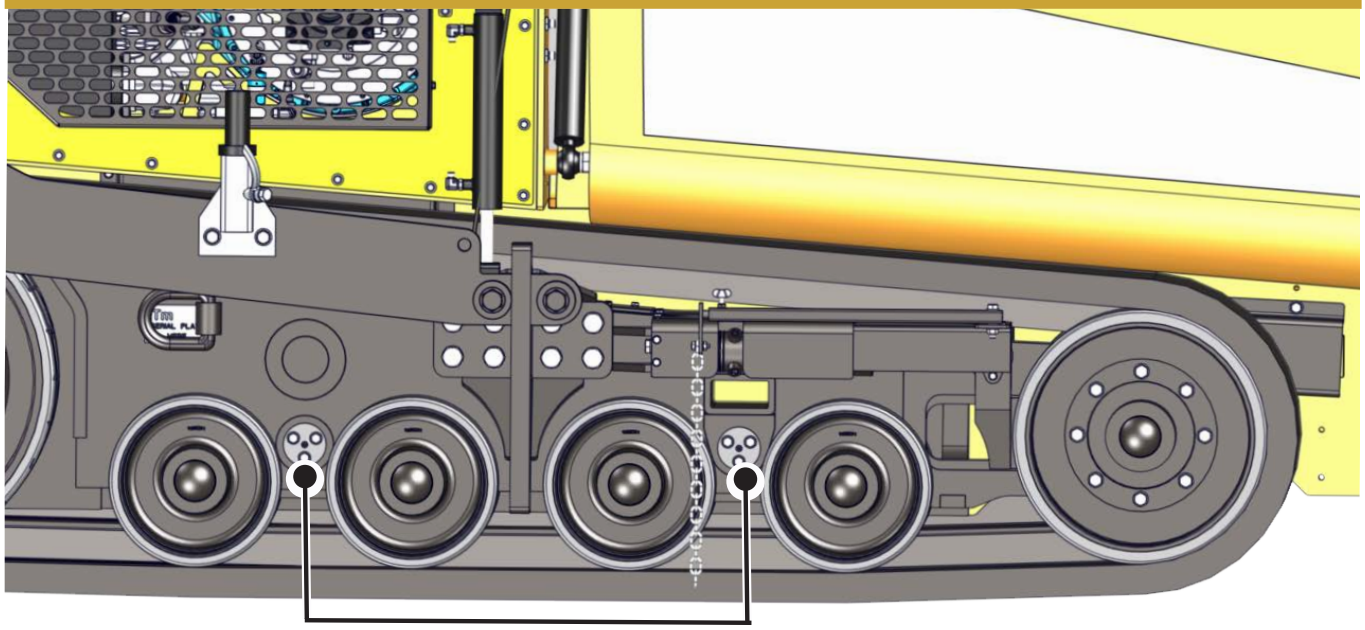
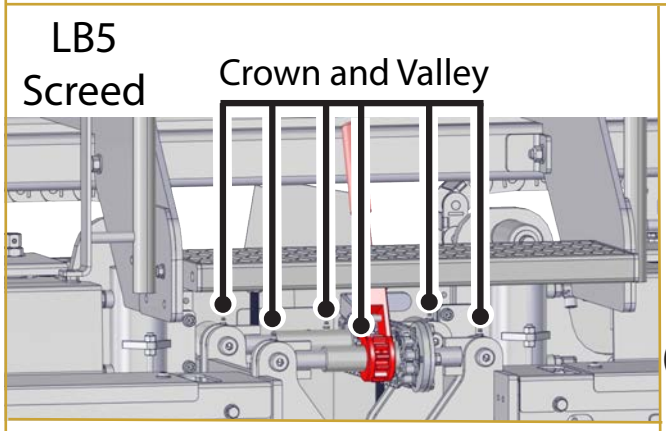
Daily Lubrication

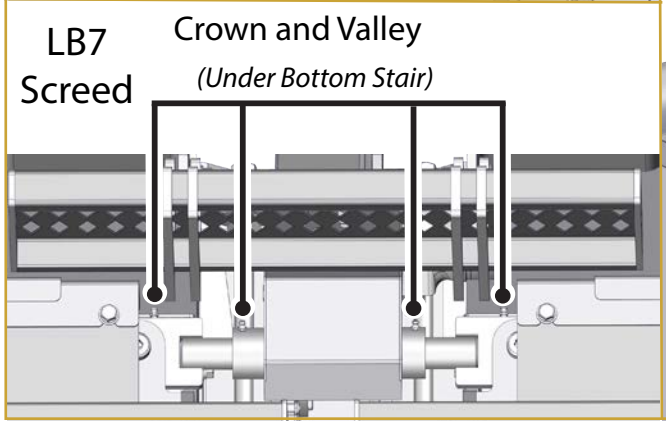


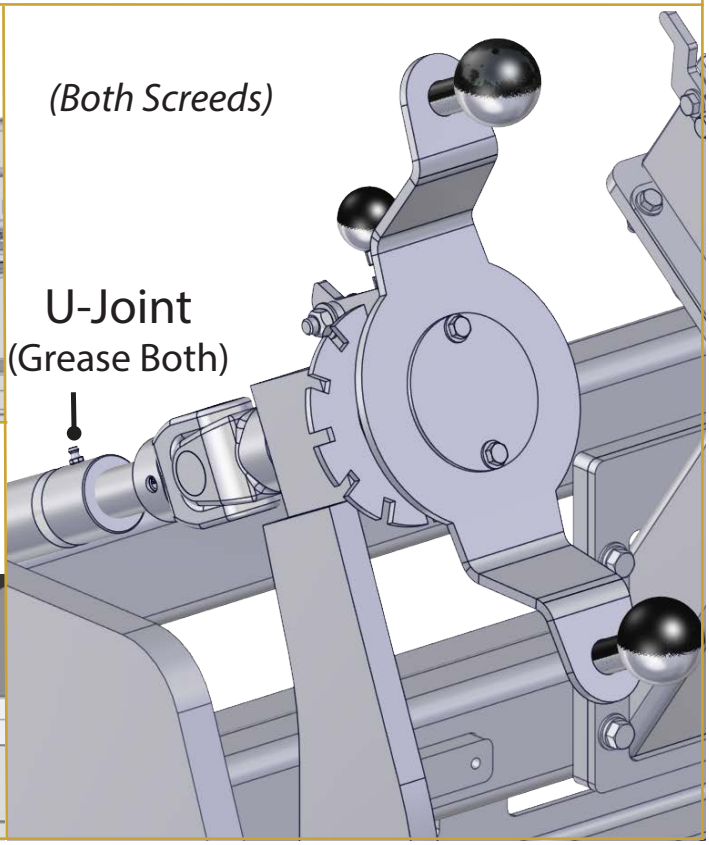
Track Pivots

Weekly Lubrication

Lubricate the screed components with Shell Avania EP Grease 2 or equivalent weekly.

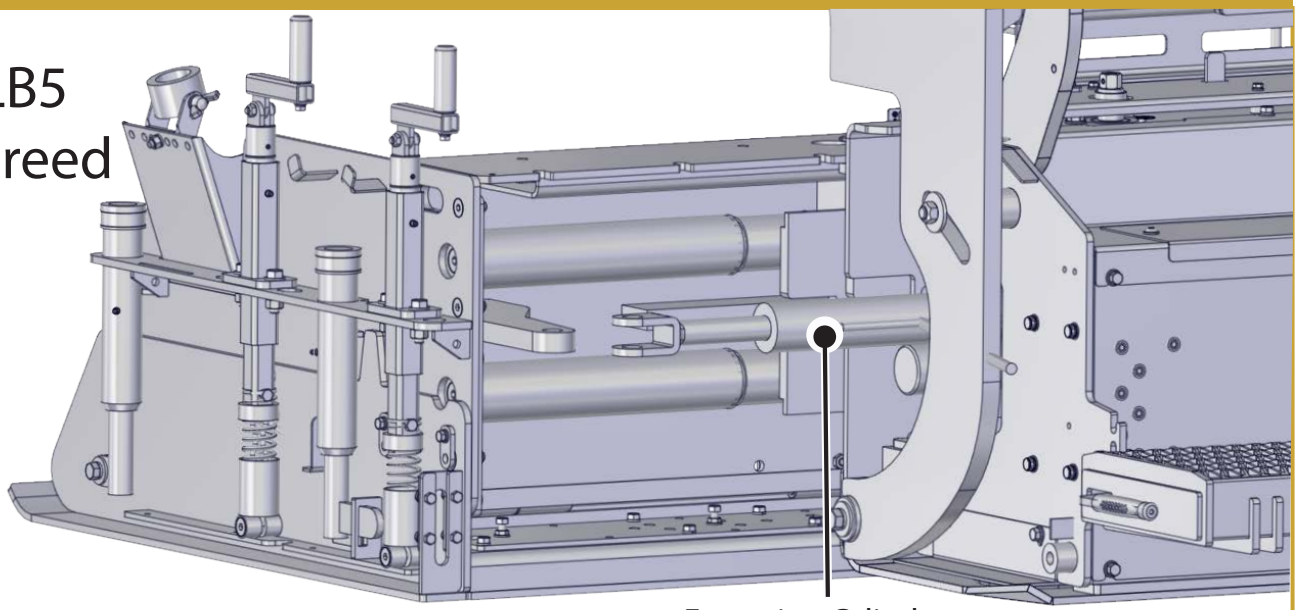






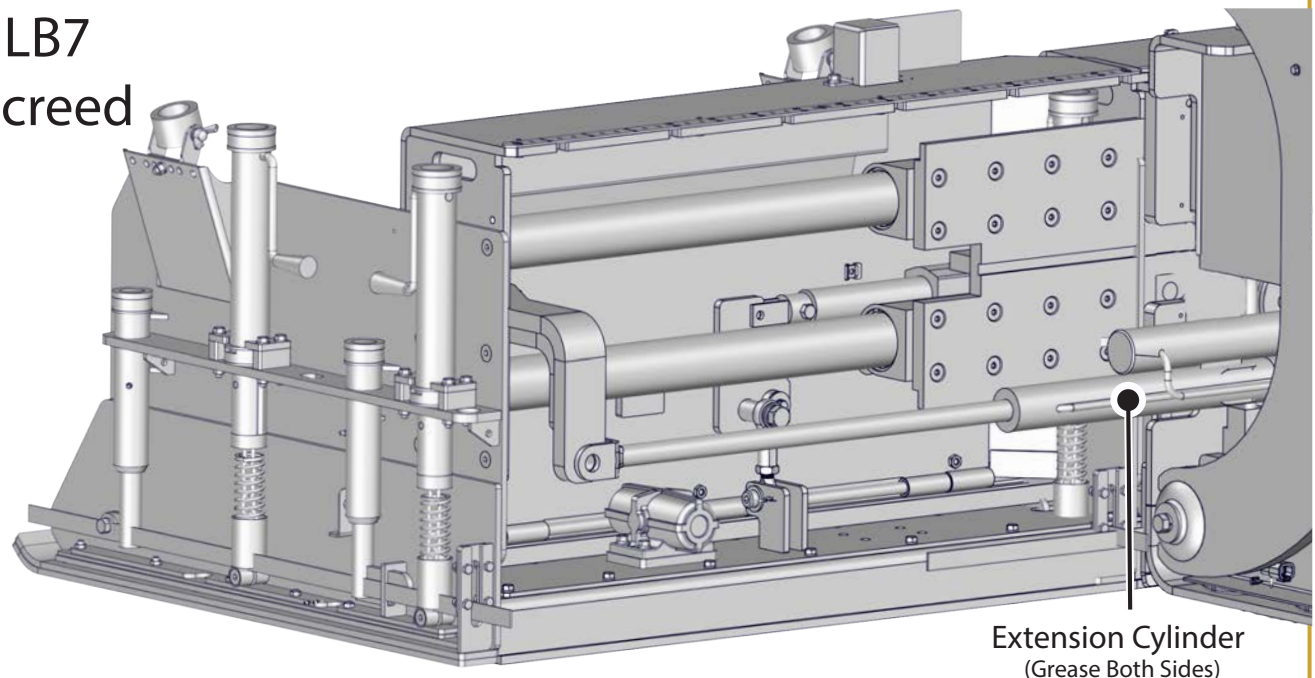
Weekly Lubrication

LB5 Screed



Extension Cylinder
(Grease Both Sides)

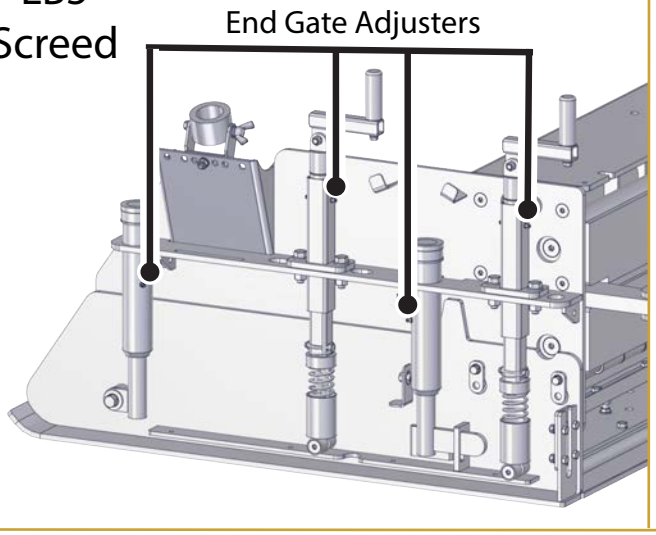
LB7 Screed



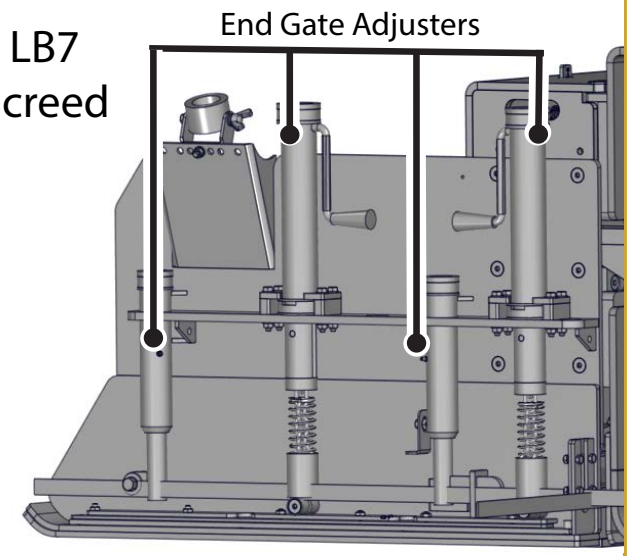
Extension Cylinder
(Grease Both Sides)

Weekly Lubrication

LB5 Screed



LB7 Screed



Torque Hub Lubrication

Check the torque hub oil level at the rear of the tracks weekly to ensure optimum operation. To check the oil level:

1. Move paver so that one of the hub plugs is in the 12 o'clock position and the other is in the 3 or 9 o'clock position as shown in **Figure 5-3.**
2. Remove the plug in the 3 (or 9) o'clock position.
3. If oil comes out, no oil is needed. Reinsert the plug and tighten.
4. If oil does not come out, remove the top plug and fill with SAE 75 W 90 synthetic gear oil until oil starts to appear at the plug.
5. Replace plug and repeat process on the other hub.



5

MAINTENANCE ADJUSTMENTS

Certain components of the paver need adjustments for optimum performance, operation and general maintenance. Keep chains clean, lubricated and properly adjusted.

⚠WARNING **Entanglement hazard! DO NOT run the engine while checking and servicing conveyor components. NEVER perform any adjustments while the engine is running.**

Conveyor Flight Chains

The conveyor flight chains must be adjusted every 100 hours. If irregular movement of the conveyor occurs, an adjustment is needed regardless of the time interval. ***(See Page 5-?? for slack tension adjustment.)***

⚠CAUTION **DO NOT adjust the conveyor flight chains too tight! Clearance (slack) is required for material to pass under and between chain rollers and the front idler sprocket. There must be *mild* tension--just enough to take up slack and prevent the chains from "knuckling."**

This chain operates most efficiently with a slight amount of slack on the return side. The chains should not:

- Be pulled too tight.
- Hang too loose.

The chains should be slightly slack with constant tension. A chain(s) "popping and cracking" indicates the chain is climbing the sprocket, which can be caused by**:**

- Excessive run-out of the front sprockets:
    - Check to verify sprocket run-out pin on the shaft. Run-out should not exceed .030 inches.
    - If excessive, replace chain or sprocket.
- Idler bearing housing brackets are not parallel to each other and the conveyor bed:
    - Check the idler bracket. If it appears twisted or misaligned, slide shaft 1-1/2 inches through the bearing inside diameter. If the centers are too far off (more than 1/4 inch), this is why the chain is climbing the sprocket.
    - If excessive, replace chain or sprocket.

Use the following procedure to make this adjustment on both sides of the conveyor:

1. Move front of paver onto a ramp so you can get underneath the front of the paver.
2. Loosen the four (4) bolts on each conveyor adjustment plate. **(Figure 5-3)**

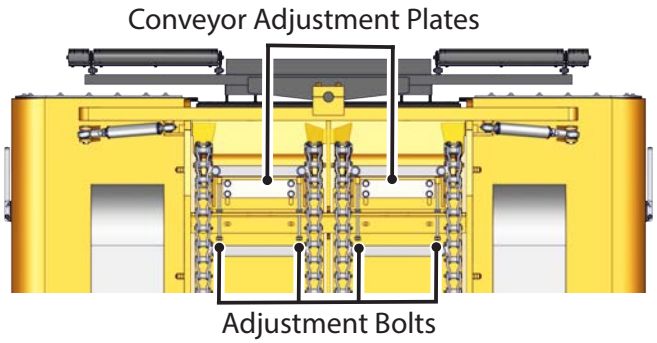


Figure 5-3. Conveyor Flight Chain Adjustment Bolts

3. Loosen the jam nuts in front of the adjustment bolts.
4. Turn each adjustment bolt alternately the same amount. For example, turn one bolt one half turn, then the other bolt one half turn, etc.
5. Continue alternating tightening until the flight chains are tight. (The pressure on the chain will be noticeable as the bolts are tightened.)
6. After the conveyor chain tension is set, retighten locknuts and bolts holding the flight chain assembly.
7. Repeat steps 1 through 4 for the opposite side.

Conveyor Drive Chains

Maintaining proper conveyor drive chain alignment is essential for conveyor performance and maximizing chain and sprocket life. Inspect the drive chain tension on each side of the paver regularly. There should only be 1/4-inch of slack on each chain. If you notice any slippage on either conveyor, the drive chain(s) may be loose. If the chains are loose, adjustment is necessary. ***(See Page 5-51 for slack tension adjustment.)***

Use the following procedure to adjust the conveyor drive chains:

1. Shut off engine.
2. Open the engine access door and locate the conveyor drive chain adjustment bolts just inside (toward rear of paver). **(Figure 5-4)**
3. Loosen the jam nuts on the chain adjuster.

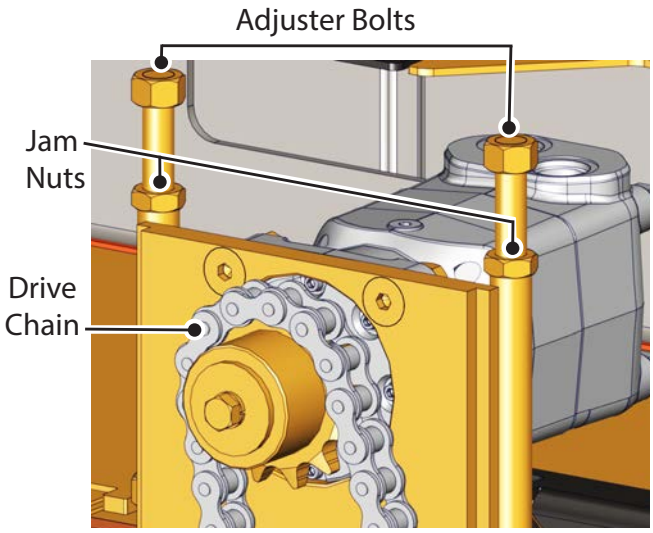


Figure 5-4. Conveyor Drive Chain Bolts

4. Turn the chain adjuster counterclockwise to loosen the chain, or clockwise to tighten the chain, leaving approximately 1/4-inch of slack. DO NOT overtighten.
5. Retighten jam nuts.
6. Repeat steps on the other side if needed.

Auger Drive Chain

The auger drive chains must also be properly aligned for optimum performance. The chains should be snug, not loose. ***(See Page 5-51 for slack tension adjustment.)*** If the chains need to be tightened, follow this procedure:

1. Loosen the lower adjuster jam nut on the right and left auger chains at the base of each auger motor. **(Figure 5-5)**
2. Turn the tension adjusters bolts counterclockwise to tighten chains.
3. Once adjusted, retighten jam nuts.

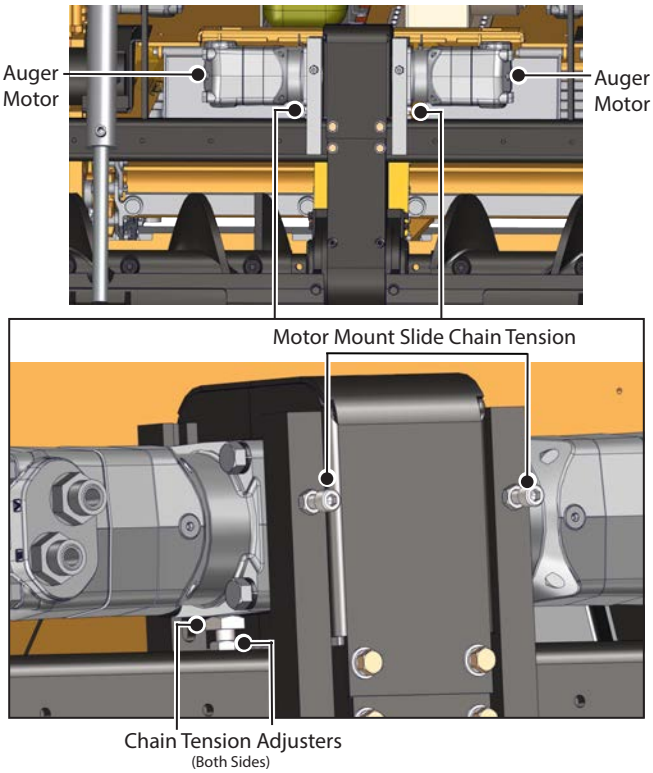


Figure 5-5. Auger Drive Chain Bolts

5

ENGINE MAINTENANCE

The LeeBoy 8680 Commercial Paver features the latest in emissions compliance and electronically-controlled engine technology. **(Figure 5-6)** Advanced emission-control devices and low-sulfur fuel requirements--only Ultra-Low Sulfur Diesel (ULSD) can be used in this engine--combine with after-treatment methods such as Exhaust Gas Recirculation (EGR), Diesel Particulate Filters (DPF), and Diesel Oxidation Catalyst (DOC). This is described in more detail on the following pages.

A common rail system optimizes combustion for a durable, quiet and fuel-efficient, turbo-charged engine. Preventive maintenance on the engine (and other paver components) is important for longevity and avoiding costly, unnecessary repairs.

Even a dirty air filter element can allow debris and dirt into the engine that can cause major problems and costly repairs in time, so be sure to follow the maintenance schedule on ***Page 5-5.*** Change oil and filters according to the maintenance schedule, keep fluids full, adjust and replace fan belt (when needed), and inspect/repair hoses to reduce wear and tear on the engine.

You can access the engine and other components via the access panels on each side of the paver or by raising the hood. ***(Page 5-6)*** The maintenance procedures on the following pages are most often required for the engine. Refer to the Engine Operator's Manual included with your paver for additional information and procedures.

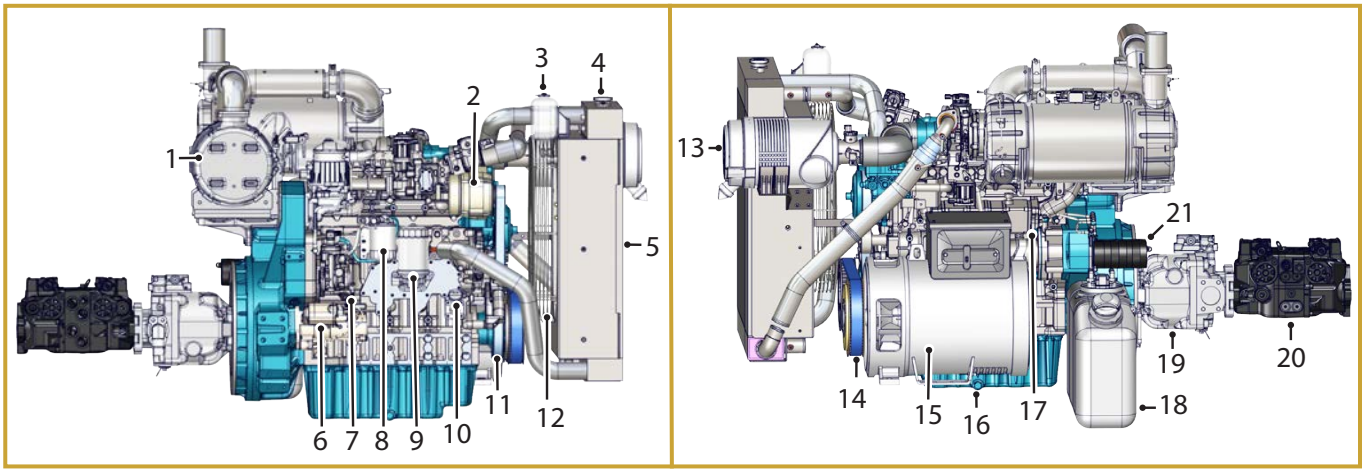


Figure 5-6. Engine and Related Components

1 - SCR (Selective Catalytic Reduction)
2 - Alternator
3 - Coolant Overflow Filler Cap
4 - Radiator Cap
5 - Radiator
6 - Starter
7 - Engine Oil Dipstick
8 - Fuel Filter
9 - Engine Oil Filter
10 - Engine Oil Filler Cap
11 - Engine Belt
12 - Cooling Fan and Safety Shield
13 - Air Filter
14 - Generator Belt
15 - Generator
16 - Engine Oil Drain Plug
17 - Auxiliary Gear Pump
18 - DEF (Diesel Exhaust Fluid) Tank/Cap
19 - Load Sense Pump
20 - Tandem Drive Pump
21 - Tandem Gear Pump

Check Engine Lubrication Oil

The engine lubrication oil must be kept at a level above the ADD mark, but not above the FULL mark, on the engine oil dipstick. **(Figure 5-7)**

⚠WARNING DO NOT check the oil while the engine is running. Hot oil can spew and cause serious personal injury.

Engine oil should be API classification CJ-4. Change the type of oil according to the ambient temperature:

Table 5-1. Ambient Temperature Oil Chart

Temperature	Engine Oil
Above 77°F (25°C)	SAE 10W-30 SAE 15W-40 SAE 30
14°F to 77°F (-10°C to 25°C)	SAE 10W-30 SAE 15W-40
Below 14°F (-10°C)	SAE 10W-30

To check the engine lubrication oil level:

1. Park the machine on a level surface and turn off the engine.
2. Wait five minutes to allow the oil to return to the oil pan so you can obtain an accurate measurement of the engine oil level.
3. Open the front hopper access panel to reach the dipstick.
4. Clean the area around the engine oil dipstick before removing it from the engine. **(Figure 5-8)**
5. Remove the dipstick from the engine and checking the oil level. The oil level should be above the ADD mark--but not above the FULL mark--on the dipstick.
6. Add oil if needed.

NOTICE DO NOT overfill. Oil can drain through the air intake system if overfilled and cause oil leaks from breather piping.

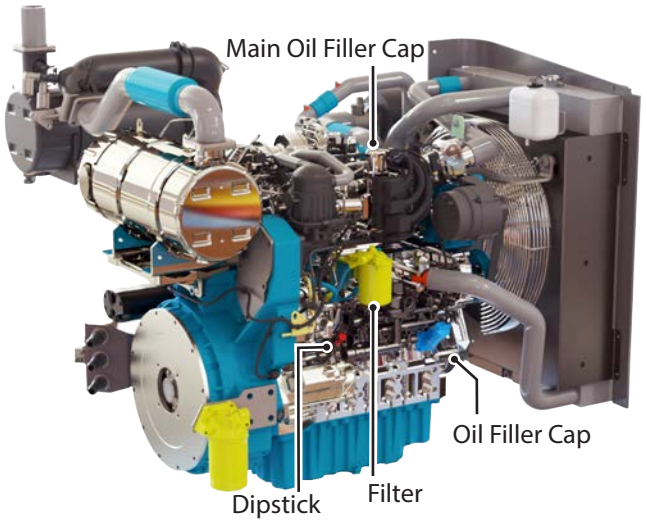


Figure 5-7. Engine Oil Dipstick

Change Engine Oil and Filter

The engine lubrication oil and oil filter must be changed at the same time. **(Figure 5-7)**

NOTE: The engine oil and filter cartridge should be changed after the initial 50 hours of operation and every 250 hour thereafter.

⚠WARNING Change the engine lubrication oil while the engine is WARM, not hot. Hot oil can cause serious personal injury.

NOTICE DO NOT change the engine lubrication oil filter while the engine is running. Serious engine damage will occur.

To change the engine oil and filter:

1. Park the paver on a level surface and turn off the engine. Allow the engine to cool down sufficiently as oil can be hot enough to cause burns.
2. Locate oil drain hose on right side of the paver.
3. Place a container large enough to hold at least five (5) gallons of oil underneath the drain hose.
4. Remove cap on end of the hose and drain all engine oil into the container.
5. Reinstall cap back onto the drain hose and reinsert hose back into place.

6. Wipe the area around the oil filter and its mounting base with a clean cloth.
7. Place the empty container under the oil filter.

NOTE: Before replacing the oil filter cartridge, loosen the cartridge bolt on top of the filter to allow air into the cartridge for complete drainage and less spillage. (Figure 5-8)

8. Use a filter removal wrench to remove the filter, turning counterclockwise.
9. Wipe inside the oil filter head using a clean, lint-free cloth.
10. Apply a very thin layer of oil on the rubber gasket area of the new filter element.
11. Install the new filter element onto the filter head by hand. When the gasket contacts the seal surface, fully tighten the filter (by hand only). Do not overtighten!
12. Fill the engine with five (5) gallons of oil at the fill cap.
13. Check oil level using the oil dipstick.
14. Dispose of used oil and oil filter (including rubber gasket) in accordance with local and federal environmental laws.

Change Air Filter

NOTICE **Never operate the engine without an air filter properly installed. Severe damage to the machine will occur.**

The air filter should be cleaned or changed every 250 hours of operation. With extensive use, the air filter may need to be changed more frequently. **(Figure 5-8)**

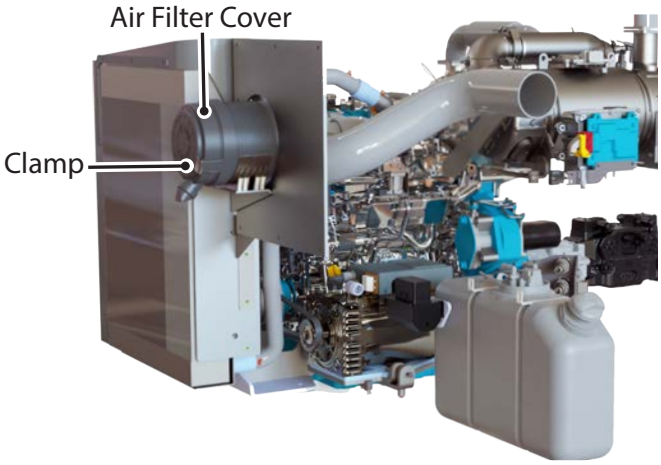


Figure 5-8. Air Filter Cover

To change the air filter elements:

1. Raise the engine hood. ***(Page 5-6)***
2. Unclamp and remove air filter cover.
3. Remove the primary and secondary air filter elements. Discard.
4. Clean the inside of the air cleaner body with a clean cloth.
5. Install new air filter elements inside the canister.
6. Reclamp cap onto the air filter cover.
7. Check clamps to ensure filter elements are tight and filter elements are making proper contact.
8. Start engine. If engine does not run smoothly, recheck filter element contact.

NOTE: The air filter elements should be replaced every 250 service hours of normal paver operation, but more often if the paver is operated in severe conditions.

Engine Drive Belt

This engine is equipped with an auto-tensioner. **(Figure 5-9)** Inspect the engine drive belt every 250 service hours and replace it every 500 hours. Regardless of the maintenance interval, the engine fan belt should be replaced if it appears frayed or cracked.

NOTE: If the fan belt is loose or damaged, and/or the fan is damaged, it can result in overheating or insufficient charging. Replace the fan belt and any other damaged components immediately.

⚠WARNING Engine must be turned off and the key removed before checking, adjusting or replacing the fan belt to avoid personal injury or even death.

To replace the belt:

1. Turn off engine and remove the key.
2. Raise the engine hood. ***(Page 5-6)***
3. Remove the safety shield from the engine cooling fan between the engine and radiator.
4. Using a 17mm socket and a ratched on the auto-tensioner pulley bolt, push counterclockwise to take tension off the belt to remove it.
5. Install new belt.
6. Reinstall cooling fan safety shield.

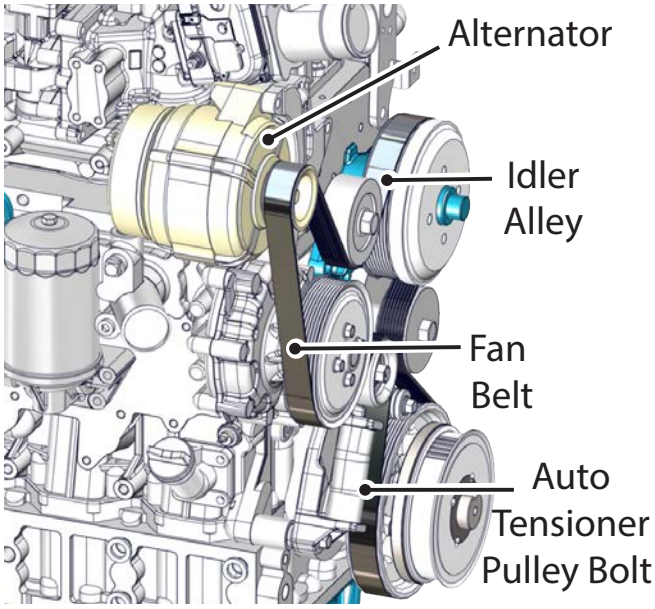


Figure 5-9. Engine Belt Tension Bolt

Change Oil Separator Element

The oil separator captures oil mist carried over from the lubrication process. The oil separator element must be changed every 1500 hours of operation. **(Figure 5-10)**

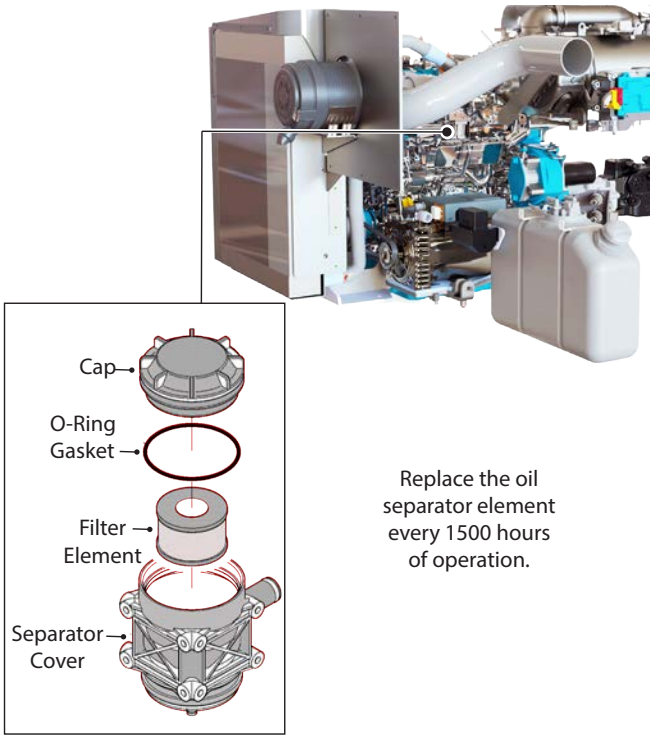


Figure 5-10. Oil Separator

To change the oil separator element:

1. Unscrew the oil separator cover cap and remove from the body.
2. Remove the used gasket and filter element. Dispose of in accordance with environmental laws and requirements.
3. Wipe oil and grease off the oil separator body and cover.
4. Insert a new filter element and gasket.
5. Reattach the cover cap to the oil separator body.

Selective Catalytic Reduction (SCR)

The SCR system is an efficient, effective means of filtering contaminates for optimum engine performance while reducing air pollution from exhaust emissions. The SCR system precisely injects Diesel Exhaust Fluid (DEF) into the engine's exhaust system to transform those toxic exhaust gases into harmless water vapor and nitrogen. SCR technology is also cost-effective and fuel-efficient. **(Figure 5-11)**

NOTICE **NEVER tamper with or remove any emission-related component under penalty of law.**

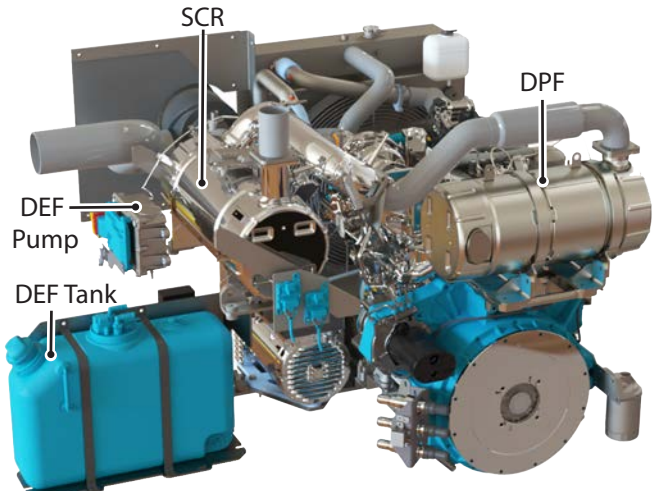


Figure 5-11. SCR System

A sensor on the Diesel Particulate Filter (DEF) tank header monitors the fluid level, temperature and quality of DEF in the tank. The header is mounted on top of the DEF tank and provides filtered DEF supply from the tank to the DEF injector through the DEF pump.

NOTE: Check the fittings around the DEF tank header for leaks or any signs of damage regularly. DEF leaks will leave a white deposit around the fittings.

⚠CAUTION **DO NOT pressure wash or steam clean this unit to avoid damaging the components. Use compressed air to remove any loose debris.**

DEF (Diesel Exhaust Fluid)

DEF is a mixture of pure urea and de-ionized water that is injected into the exhaust stream by the SCR system. Temperature and humidity affect DEF consumption, with hot and dry climates resulting in greater DEF consumption. Typically, machines will consume an amount equal to 2 - 10 percent of the fuel you burn.

The DEF tank is located just inside the right engine access door. **(Figure 5-12)** DEF tanks always have a blue cap with a smaller fill neck to prevent erroneously filling with the wrong fluid.



Figure 5-12. DEF Tank (Blue) Fill Cap

DEF fluid is available at most truck stops, diesel engine dealerships, auto supply stores and large gas stations. To find DEF locations in the U.S., go to http://www.finddef.com/def_map.php.

⚠WARNING **DEF contains urea, therefore, wear eye protection, gloves and appropriate protective clothing when filling the tank and handling DEF. Avoid skin contact. Use only with adequate ventilation and avoid breathing vapor or mist. DO NOT get the substance in your eyes. In case of contact, immediately flush eyes for a minimum of 15 minutes. ALWAYS wash hands thoroughly after handling DEF.**

NOTICE **DO NOT use DEF after its expiration date. Always dispose of DEF in compliance with federal and local environmental laws.**

It is critical that you prevent contamination entering into the SCR system. Even trace amounts of contaminants introduced into the SCR system can cause damage.

- **Always** clean around the fill neck and dispensing nozzle when refilling the DEF tank. If you're refilling the DEF tank often, the opportunity for airborne dust and dirt ingestion is high.

- Be especially careful handling DEF at the job site. While it's a good idea to have DEF on hand, especially in rural areas, the risk of contamination is higher. Be sure the ambient temperature is within the correct storage range if bringing DEF to the work site.

DEF Storage

DEF is a vital element in engine performance. Proper storage and use will help to insure maximum engine performance and a long, trouble-free experience with the installed engine.

DEF is a mixture of pure urea and de-ionized water that is injected into the exhaust stream by the SCR system. Temperature and humidity affect DEF consumption, with hot and dry climates resulting in greater DEF consumption. Temperatures also affect the quality of DEF and can impeded peak performance.

- **DEF freezes at 12°F.** In winter months it should be stored above this temperature. If DEF does freeze in the machine tank during overnight storage between jobs, onboard systems enter into Thaw Control Mode to heat the DEF back to usable temperatures. Engine maximum torque may be limited while DEF is frozen.
- **Temperatures above 86°F causes DEF to break down**, which releases ammonia into the onboard DEF tank or storage container. This breakdown of the DEF affects the quality of the DEF, which could yield warning messages or reduce engine performance of the machine, and may increase pressures within the storage containers. During DEF fill, or after any prolonged storage of DEF, if the smell of ammonia is present, the DEF fluid has started to break down and LeeBoy recommends it be changed or discarded.

If DEF is stored or used above 86°F, the DEF life will be reduced. DEF storage recommendations include:

- Store DEF in a clean, dry, well-sealed container in a well-ventilated location away from dust and direct sunlight.
- Keep DEF away from any container of gasoline and diesel fuel or other lubricants.
- Store DEF in a dedicated container that has not been used with other fluids, including funnels and hoses.
- If using pumps to fill DEF tanks, ensure the seals, O-rings and fittings on the dispensers are DEF-compatible.
- The best practice is to pour directly from the DEF container into DEF tank.
- Use a container size that allows DEF consumption in 2 months.

Other best practices for the use of DEF fluid:

- Fill DEF tank with every fuel fill to maintain fresh high quality DEF within the system.
- Avoid prolonged storage of DEF within the parked machine.
- Drain DEF fluid during prolonged storage.
- Drain DEF fluid during winter months when DEF may freeze for prolonged periods.

NOTICE **At 80°F, DEF has a storage life of one (1) year.**

DPF Service

The DPF is designed to burn trapped particulate matter automatically or via operator-induced regeneration. The longer the engine runs, the more ash (burnt residue) is collected on the filter. Too much residue adversely affects DPF performance, therefore, see your authorized dealer for professional cleaning (or replacement) if needed.

Do not be alarmed if you notice white exhaust from the exhaust pipe at start-up or when speeding up. White exhaust or water may come out during cold weather or after long-hour idling due to condensation.

If DPF regeneration occurs frequently, i.e., five hours or shorter, on a routine basis, change the oil. If this condition continues after an oil change, contact your authorized dealer for service. (Refer to your Engine Operator's Manual for service information.)

NOTICE **Too much idling can cause DEF to crystallize in the injectors if the machine isn't hot enough. Don't idle machine any longer than necessary as too much idling can cause soot to build up in the Diesel Particulate Filters (DPF), requiring more frequent regeneration.**

NOTICE **To reduce the amount of soot and maximize DPF efficiency, use API-CJ4 (or ACEA-E9 European equivalent) engine oil. Running the engine at 1800 RPMs will also provide more efficient operation and allow automatic regeneration more efficiently while paving.**

COOLING SYSTEM

Inspect the radiator and oil cooler daily. Use a low-pressure water hose to clean as needed once the engine has cooled down.

Inspect the coolant system daily. Wash out the radiator with a low-pressure water hose ONLY when the engine is cool.

Check radiator hoses and clamps every six months for leaks or damage, ensuring hose clamps are securely tightened. If radiator hoses are swollen, hardened or cracked, replace them immediately.

⚠CAUTION Inspect the radiator and oil cooler daily. Wash out as needed with low-pressure water hose after the engine has cooled down. Plugged or dirty radiators and oil coolers cause engine or hydraulic overheating.

⚠CAUTION Replace radiator hoses and clamps every two years, or sooner if needed. Damaged radiators hoses or coolant leaks also cause overheating.

If the coolant temperature warning icon illuminates, or steam emits from the radiator or overflow pipe, turn off the load but **idle the engine at least five minutes** to cool down gradually and check the following items:

- Inspect for any coolant leaks, obstacles or debris around the air inlet or outlet, dirt or dust between radiator fins and tube, or clogged water pipe.
- Check the fan belt as described in previous section.
- Check that radiator cap is properly tightened. A gap between the cap and seal accelerates coolant loss.
- Check for any internal blockage in the radiator hose. If scale forms on the hose, clean with scala inhibitor (or equivalent).
- Check engine cooling fan for damaged or bent blades. Replace if damaged.

Replace Radiator Hoses

Radiator hoses should be replaced if worn or frayed. Access the radiator from the top access panel.

NOTE: Inspect the radiator and oil coolers daily. If needed, wash out using a low-pressure water hose once the engine has cooled down.

⚠CAUTION Dirty or plugged radiators and oil coolers will cause engine or hydraulic overheat conditions.

⚠WARNING This procedure should only be performed when the machine is not running.

⚠WARNING DO NOT remove the radiator cap while the engine is still hot. Verify there is no pressure in the radiator BEFORE removing the radiator cap. Serious burn injuries can occur if cap is removed while hot and under pressure.

1. Turn off ignition, remove key, and allow the engine to cool down.
2. Raise the engine hood. ***(Page 5-6)***
3. Carefully remove the radiator coolant fill cap. **(Figure 5-13)**

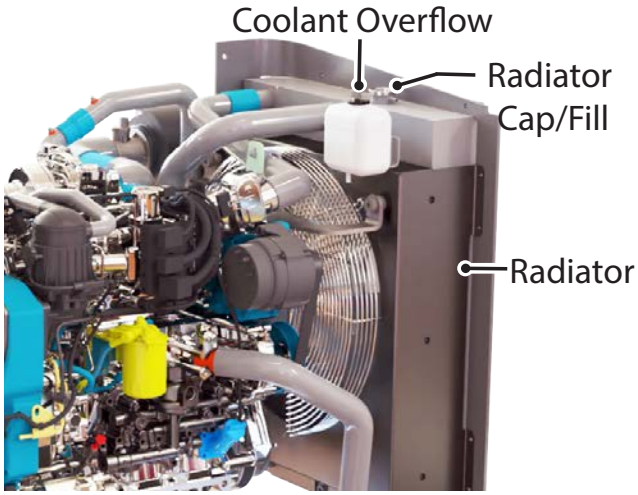


Figure 5-13. Radiator and Coolant Fluid Caps

4. Locate the radiator drain hose (with petcock) inside the left engine panel. Drain radiator fluid into a container large enough to hold seven (7) gallons.
5. Remove the upper and lower hoses by loosening the hose clamps on each.
6. Disconnect hoses and inspect for damage.
   - If clamps are not damaged, retain for reinstallment (or replace if damaged).
7. Install new hoses and reconnect with hose clamps.

FUEL SYSTEM

Fuel Tank

The fuel level is indicated on the digital display fuel gauge. Use only low-sulfer diesel fuel and always fill the fuel tank at the end of the work day to prevent condensation. The fuel tank has a green cap and is located inside the left access panel of the paver. **(Figure 5-14)**

NOTICE **Fill up with low-sulfer diesel fuel at the end of the day to reduce condensation in the tank.**

⚠WARNING **Diesel fuel is highly FLAMMABLE. DO NOT smoke while filling the fuel tank. Fill the fuel tank in a designated area that provides proper ventilation with a fire extinguisher available.**

⚠WARNING **Explosion Hazard! Never fill the fuel tank near an open flame, or near equipment that can create sparks. Never check fuel level or fuel system components near an open flame.**

⚠WARNING **Explosion Hazard! Stop the engine immediately if any fuel leakage is noted. DO NOT start the engine until the leak has been corrected.**

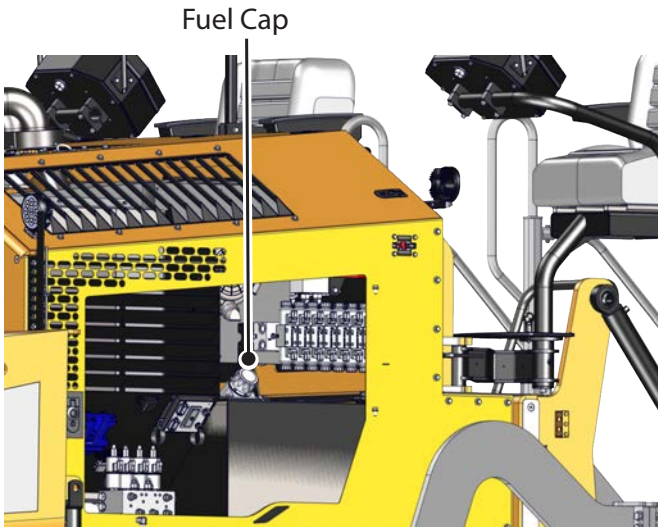


Figure 5-14. Fuel Cap

To fill the fuel tank:

1. Stop engine.
2. Remove fuel cap from fuel tank.
3. Fill with diesel fuel until full.
4. Replace fuel cap.

Change Fuel Filter

The fuel filter, water separator and fuel pump with bowl filter are conveniently located together on the engine access door in the hopper. **(Figure 5-15)**

⚠WARNING **DO NOT change the fuel filter while the engine is running. DO NOT smoke or change fuel filter near open flames as diesel fuel is highly combustible. Be careful not to spill fuel.**

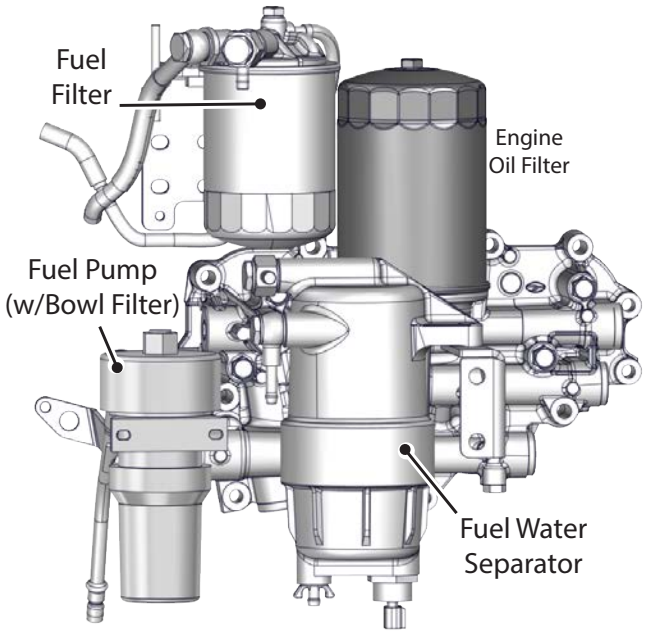


Figure 5-15. Fuel Filters

Replace the fuel filter using the following general procedure:

1. Stop the engine and allow it to cool. Put a container under the filter before removing it.
2. Use a filter removal wrench to loosen and remove the filter by turning it clockwise.
3. Drain and discard the removed filter in compliance with local and federal environmental laws..
4. Wipe a little fuel on the rubber gasket. Install the new fuel filter element onto the filter head, **carefully tightening the element by hand. DO NOT overtighten.**
5. Turn the key to the ON position and allow the fuel pump to fill the filter, assisting with the primer bulb.
6. Start the engine and check for any fuel leaks. **Stop the engine immediately if any leakage is noted. DO NOT start the engine again until the leak has been corrected.**

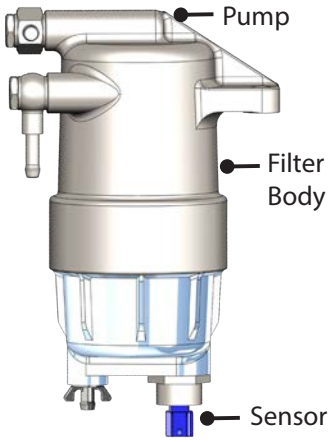
Fuel Water Separator

Water and dirt that gets into fuel settles into the water separator. **(Figure 5-16)** An error code will appear on the DM430 digital display when it needs to be emptied. It is important to drain it immediately when this occurs. The fuel water separator is located on the access door in the hopper.

NOTICE **If you run out of fuel, it is important to pump the water separator to prime fuel injection lines. Failure to do so can adversely affect performance.**

To drain the water separator:

1. Pull retainer clip from plug to disconnect the harness.
2. Unplug the harness, then unscrew the plug.
3. Drain and reinsert components.



To clean the water separator:

1. Follow the previous steps for draining the water separator.
2. Loosen and remove the filter body, then clean it with light oil.
3. Reinsert and tighten.

Figure 5-16. Fuel Water Separator

HYDRAULIC SYSTEM

Check Hydraulic Oil

The hydraulic motors and hydraulic cylinders use the same hydraulic oil tank and hydraulic oil supply.

NOTICE **Take care when working with the hydraulic system to ensure it is completely clean.**

NOTICE **Never allow the hydraulic oil level to fall too low. Pump damage will occur.**

⚠WARNING **DO NOT use your hands on any hydraulic hoses, fittings or system component when checking for leaks. Serious personal injury can result from an oil leak under high pressure. Oil can be injected under the skin by high pressure. Protect the eyes by wearing safety glasses.**

⚠WARNING **DO NOT loosen or remove the hydraulic oil strainer fill cap when the hydraulic oil is hot. Always loosen the cap slowly to relieve any pressure in the hydraulic oil reservoir.**

Check the hydraulic tank oil level daily using the sight gauge just inside the right engine access panel. **(Figure 5-17)** Add hydraulic oil if needed (the hydraulic oil cap is just above the sight gauge):

- Fill to the black line on decal to the right of the sight gauge if the hydraulic oil is cold.
- Fill to the fill line if the hydraulic oil is at normal operating temperature (preferred method for an accurate reading), shown at **Figure 5-17**.

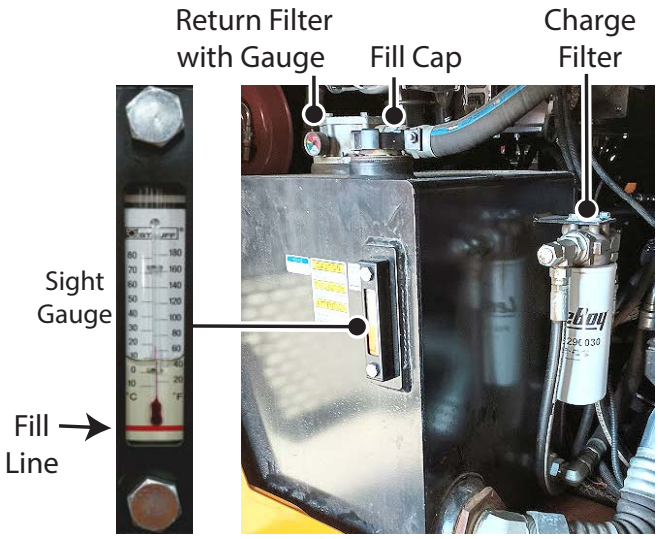


Figure 5-17. Hydraulic Oil Tank, Cap, Filters and Sight Gauge

- Slowly unscrew the hydraulic oil strainer fill cap (above sight gauge) and add hydraulic oil.

NOTE: DO NOT use unfiltered hydraulic oil.

- Check the hydraulic oil level using the sight gauge.
- Keep the hydraulic oil filled to the correct level.

NOTICE DO NOT overfill the hydraulic oil reservoir.

NOTICE Leeboy uses a leak-detection dye that may turn the hydraulic oil a slight yellow tint.

Change Hydraulic Oil and Filters

Changing the hydraulic oil removes the accumulation of dirt, water and mechanical wear particles from the hydraulic oil system. Clean, filtered oil is critical to optimize hydraulic system operation. The hydraulic oil system holds approximately 37 gallons.

There are two hydraulic filters. The hydraulic charge filter is located inside the right engine access panel on the right side of the hydraulic oil tank. **(Figure 5-18)** The hydraulic return filter is located

NOTICE Contaminated hydraulic oil can shorten the expected service life of components in the hydraulic system.

⚠WARNING DO NOT use your hands on any hydraulic hoses, fittings or system component when checking for leaks. Serious personal injury can result from an oil leak under high pressure. Oil can be injected under the skin by high pressure. Protect the eyes by wearing safety glasses.

Use the following procedure to change the hydraulic oil and filter:

1. Stop the engine.
2. Allow the hydraulic oil to cool, until it is at a warm temperature only.
3. Slowly remove the hydraulic oil strainer fill cap. Do not remove the strainer portion under the cap.
4. Place a clean, lint-free cloth over the reservoir fill tube opening and secure in place with tape.

⚠WARNING DO NOT drain the hydraulic oil from the tank when it is hot as the oil can cause serious injury. Drain at a warm temperature only.

5. Place a container with sufficient capacity to collect the hydraulic oil under the hydraulic tank plug.
6. Carefully remove the plug from the bottom of the hydraulic tank and allow all the hydraulic oil to drain into the container.
7. Reinstall the hydraulic oil drain plug.
8. Change the hydraulic oil filter by turning the filter counterclockwise, then replace with new filter turning clockwise until snug (by hand only). **(Figure 5-18)**
9. Remove the strainer inside the oil filler cap and replace with a new one, aligning the holes on the strainer with the mounting holes on the gasket.
10. Fill the hydraulic oil tank with filtered hydraulic oil until proper level is shown on the sight gauge.

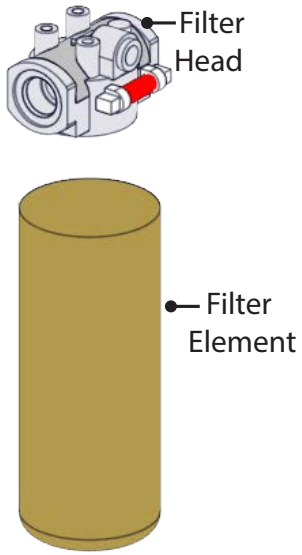


Figure 5-18. Hydraulic Charge Filter

11. Check the oil level in the hydraulic oil tank again. Add oil if needed.
12. Reinstall the hydraulic oil cap onto the reservoir filler neck and tighten securely.
13. Start the engine.
14. Check the hydraulic system for any leaks.

⚠CAUTION Stop the engine immediately if any hydraulic leak is noted. DO NOT start the engine until any problem noted has been corrected.

5

ELECTRICAL SYSTEM

The electrical system is a 12-volt negative ground system utilizing two batteries. Raise the hood to access the batteries directly in front of the right operator platform. **(Figure 5-19)**



Figure 5-19. Batteries

Connect the positive (+) cable to the positive (+) post of the battery. Connect the ground cable to the negative (-) post of the battery. It is advisable to disconnect the negative (-) cable first and connect it last. Reversed polarity can damage the electrical system.

⚠WARNING Fire and Explosion Hazard! Always observe battery polarity when connecting a battery charger or jumper cables to the battery: negative (-) to negative (-), positive (+) to positive (+). Failure to do so can produce sparks.

When connecting a booster battery, connect one end of the first jumper cable to the positive (+) terminal of the dead battery and the other end to the positive (+) terminal of the booster battery. Connect one end of the second jumper cable to the negative (-) terminal of the booster battery and the other end to the frame of the paver with the dead battery.

⚠WARNING Fire and Explosion Hazard! Be sure that the battery charger is in the OFF position before connecting it to the battery.

Be sure to keep the battery fully charged during cold weather to keep it from freezing. Freezing weather has little effect on a fully-charged battery.

NOTICE ALWAYS turn the master battery switch off when working on the electrical system or welding anywhere on the machine. Damage to electrical components could result.

NOTICE DO NOT pressure-wash electrical components, especially around the generator. Protect electrical components when washing paver.

Battery Servicing

⚠WARNING Batteries contain sulfuric acid. ALWAYS wear safety goggles and protective clothing when servicing the battery.

⚠WARNING NEVER allow battery fluid to come into contact with clothing, skin or eyes. If battery fluid contacts the eyes or skin, immediately flush with clean water and obtain prompt medical treatment.

⚠WARNING Keep sparks and flames away from the batteries as electrolyte gas is highly flammable.

1. Raise the hood to access the batteries. ***(Page 5-6)***

NOTE: When replacing the battery, discard the old battery properly.

2. Before connecting the batteries, turn off the master switch, located in the right access panel at the front of the operator station. **(Figure 5-20)**

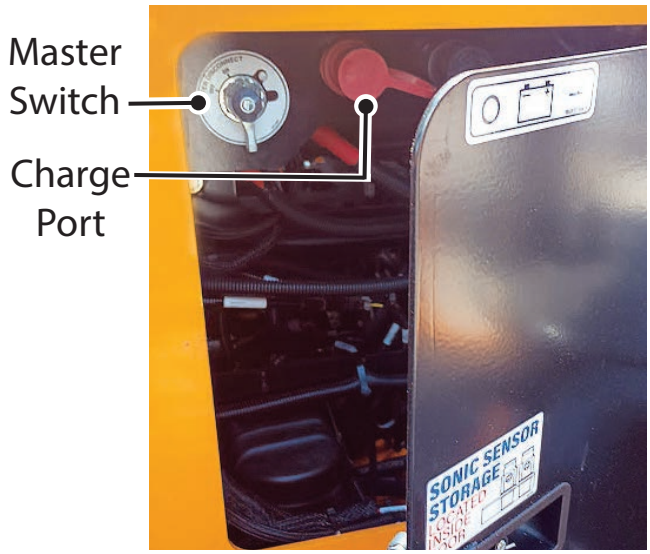


Figure 5-20. Battery Master Switch

3. Be certain that battery terminals and posts are clean and that the battery cable terminals are tight.

NOTICE Dirty or loose battery connections can create high electrical resistance and permit arcing.

4. If corrosion is present around terminal connections, remove and wash with ammonia or a solution of 1/4-pound baking soda added to one quart of warm water. Ensure the vent caps are tight to prevent solution from entering the cells.
5. Pour clean water over the battery and surrounding area to rinse away the solution.
6. Check vent cap breather openings to ensure they are clear.

Alternator

The alternator supplies electrical power to electronic controls and charges the battery. Its built-in regulator controls voltage output. **(Figure 5-21)**

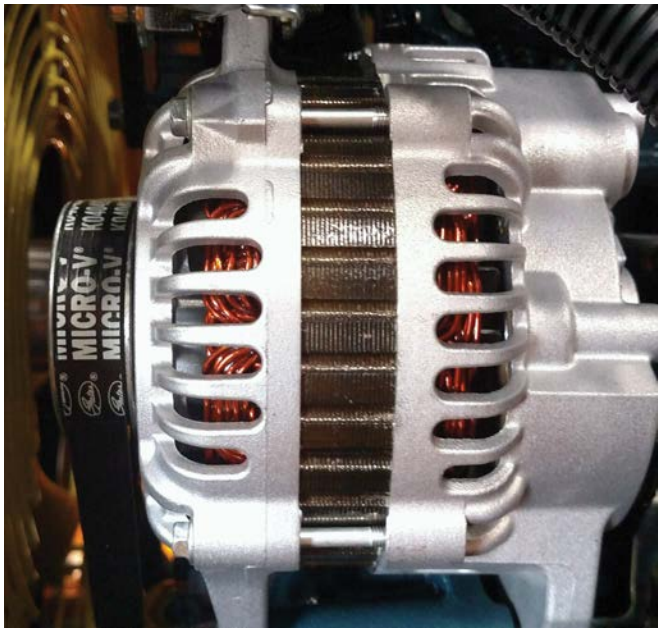


Figure 5-21. Alternator

Use the following precautions to prevent damage to the alternator or regulator:

- Never polarize the alternator.
- Never ground any alternator terminals or circuits.
- Always disconnect the battery before disconnecting or connecting the alternator.
- Never disconnect the alternator while machine is operating.
- Ensure wiring is properly connected before connecting the battery.
- Always connect a booster battery in the proper polarity: Negative (-) to negative (-) and positive (+) to positive (+).

Adjust Generator Belt

Adjusting the generator belt every 250 hours (quarterly) for optimum performance is a simple procedure. You can access the generator from the center engine access panel or by raising the hood. ***(Page 5-6)***

NOTICE **Be sure to maintain the belt alignment during and after adjustment.**

The key to adjusting the generator belt is loosening and/or tightening all four adjustment bolts to align the generator pulley and tension the belt:

1. ***Slightly*** loosen the four bolts on the generator plate. **(Figure 5-22)**
2. ***Slightly*** tighten each mount bolt until you have approximately 1/2 to 3/4 inches deflection with minimal force.
3. Retighten the bolts on the generator plate.

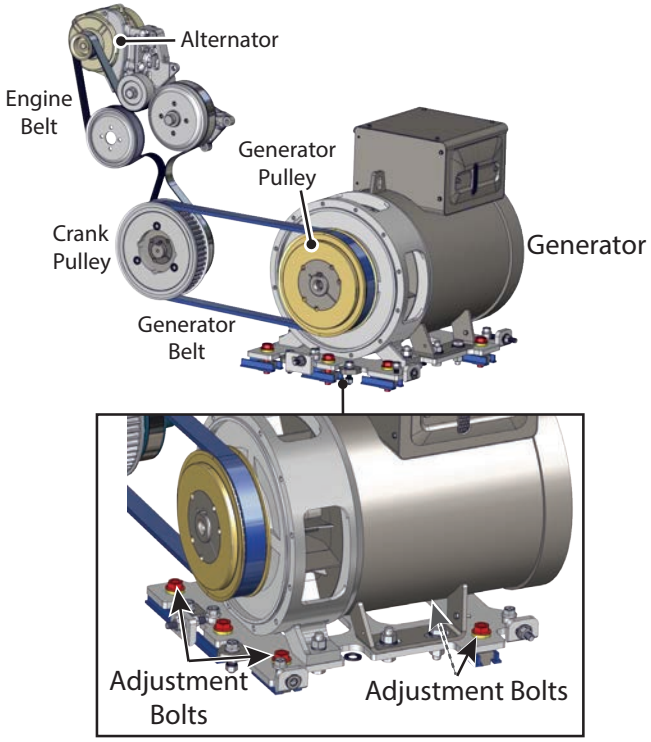


Figure 5-22. Generator Belt Adjustment

5

Fuse Panel

The electrical system is protected from overload damage by fuses. If an electrical component fails to operate, check the fuse box panel first to ensure that a fuse has not "blown." A blown fuse is an indication that there is an overload that needs to be corrected.

The fuse panel is located inside the junction box for convenient access. **(Figure 5-23)**

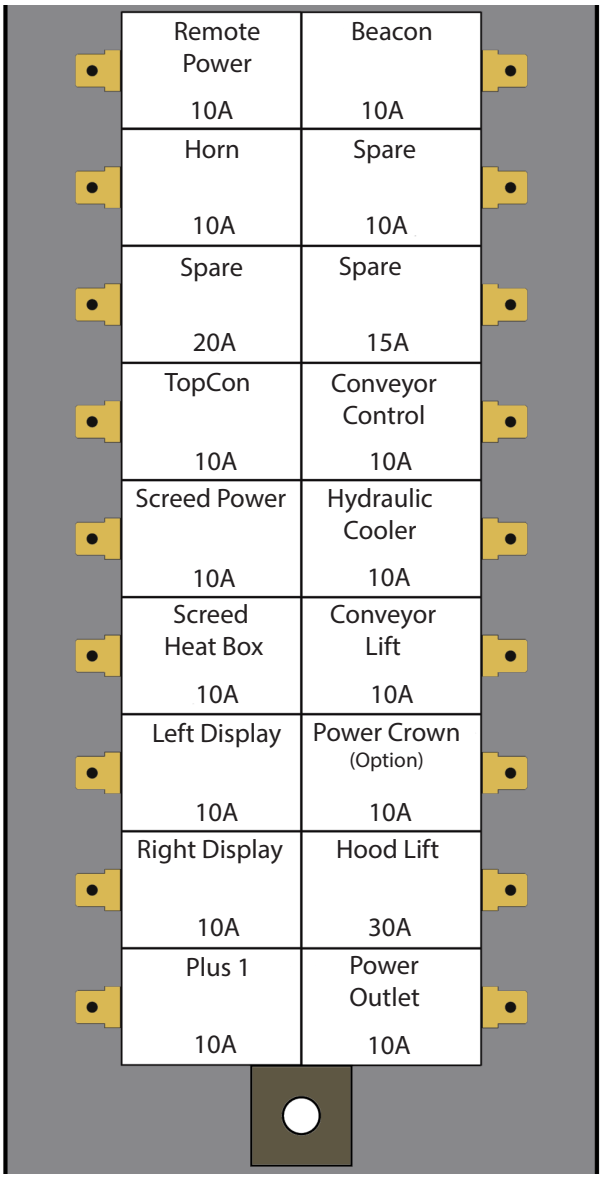


Figure 5-23. Fuse Panel

Electric Heat Control Box

The electric heat control box contains the element breakers and main outputs for the screed heating system on both the standard LB5 and optional LB7 screeds. **(Figures 5-24 and 5-25 *on the following page.)***

The standard LB5 screed is powered at 240 VAC while the optional LB7 screed is powered at 480 VAC. Heating element relays "make" or "break" the circuit to each element to start or stop the heating cycles. When the HEAT ON button is depressed, 12VDC is momentarily applied to the main timer relay to start the timing cycle.

NOTE: Control boxes are manufactured to fit all screed and paver combinations. If not all plugs are connected to wires, it may be normal. Be sure to keep unused plugs covered with mating protective caps.

DANGER The heat box is high voltage ! ALWAYS be careful when working with electrical components to avoid electrical shock. Take extra precautions to avoid electrical shock that could cause serious injury or even death.

- Each element output consists of two wires--one connects to the L-1 circuit and the other to the L-2 circuit.
- Each breaker has two terminals--one is connected to the main input and the other terminal to an element output wire.
- The L-1 circuit is the left bank of element breakers.
- The L-2 circuit is the right bank of element breakers.

Heating element relays "make" or "break" the circuit to each element to start or stop the heating cycles. When the HEAT ON button is depressed, 12VDC is momentarily applied to the main timer relay to start the timing cycle.

LB 5 Screed Heat Box

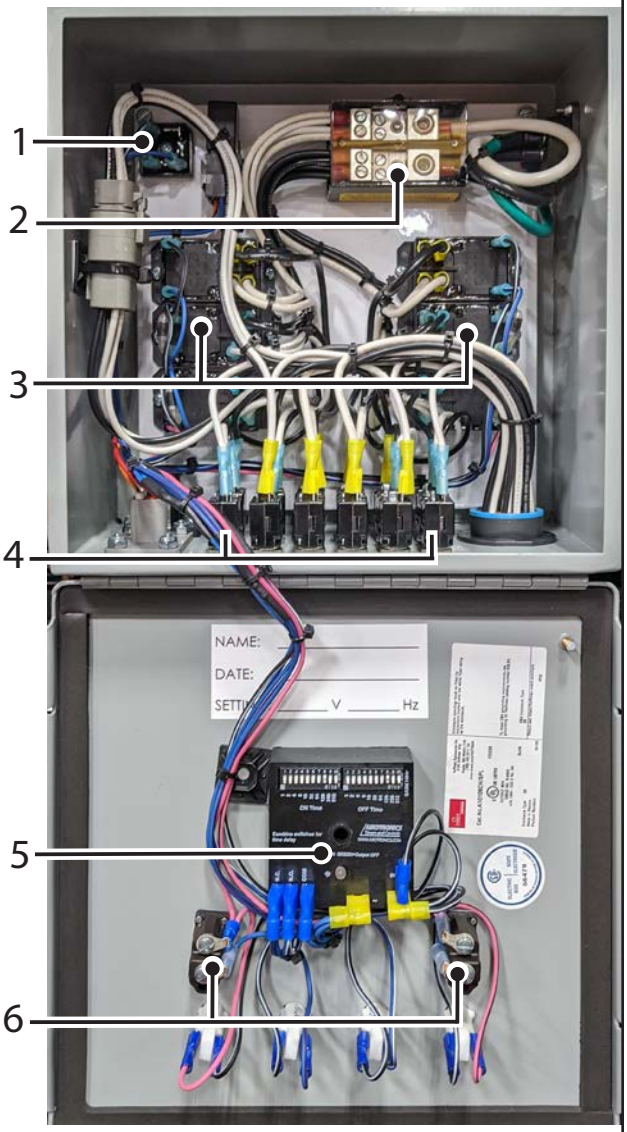


Figure 5-24. Electric Heat Elements

1 - Rectifier

2 - Terminal Block

3 - Heating Element Relays

4 - Element Circuit Breakers

5 - Cycling Timer

6 - Toggle for Left/Right End Gates

LB 7 Screed Heat Box

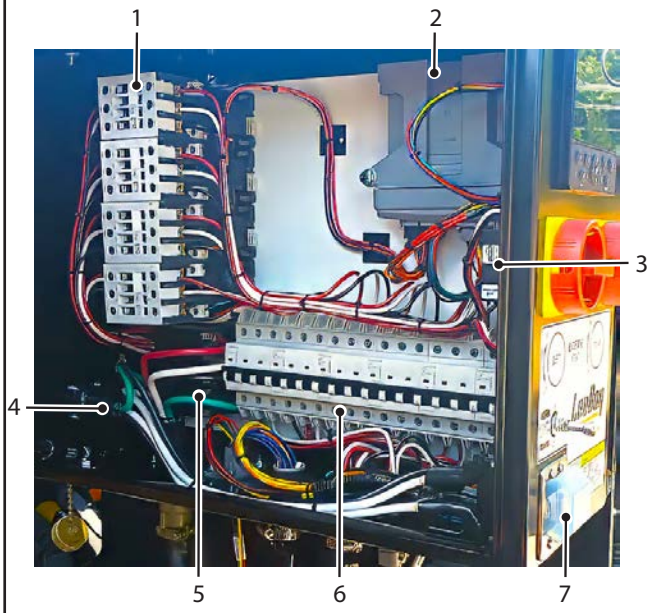


Figure 5-25. LB7 Screed Electric Heat Elements

1 - Heating Element Contactors

2 - OX Module

3 - Switches for Left/Right End Gates

4 - 110 VAC Connector (OPTION)

5 - Terminal Block

6 - Element Circuit Breakers

7 - 110 VAC Circuit Breaker (OPTION)

5

Heating Elements

Each heating element is sized to fit properly and provide sufficient heat to the screed plate. Elevated temperature prevents mix from sticking to the lower surface of the screed plate.

⚠ DANGER DO NOT operate an element with a known electrical shortage. Replace faulty elements and wiring immediately.

An element assembly consists of four main components:

- The element.
- The wire protector adapter.
- The wire protector.
- The four-pin wire plug at the end of the element protector.

Each element is covered by a thin strip of insulation to keep heat from escaping. A support bar and shield protects the element assembly.

Each element is clamped onto the screed plate to provide a positive and efficient connection. Enough pressure should be applied to the element assembly to sufficiently hold it securely against the screed plate surface.

To remove an element:

1. Remove top cover from the extension screed plate.
2. Loosen all the clamping studs over the element.
3. Remove and replace element.
4. After tightening the clamping stud, lock the clamp by tightening the stud jam nut.

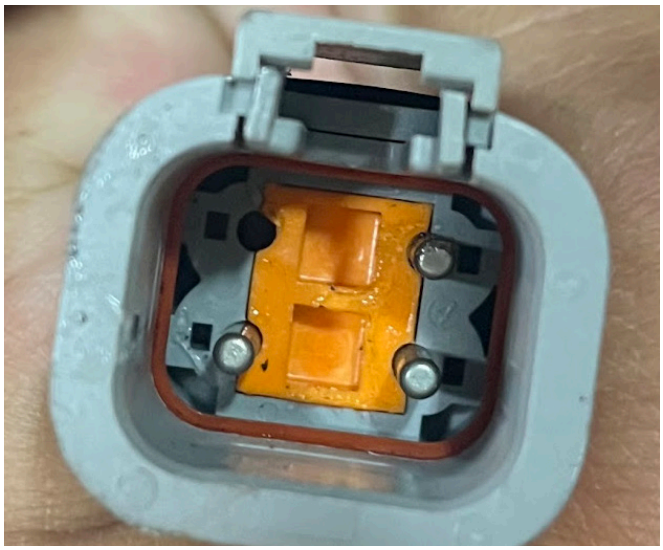


Figure 5-26. 4-Pin Plug

Element Resistance Testing

The breakers are wired into each leg of each element. If an element has a fault, either in the wiring, or in the element itself, the breaker will trip and power will no longer be applied to that leg of the element.

When a breaker in the control box has “tripped,” there may be a problem in the wiring or an element in the circuit.

The breakers can be manually reset by depressing the trip button back into place when they are extended. If the breaker still does not reset, you need to test or possibly replace an element.

If the element is functioning correctly, you should read a resistance between the connector pins when testing. A faulty element will show high resistance, indicating a bad element. ***(Refer to* Tables 5-2 and 5-3 *on the following page.)***

⚠ DANGER HIGH VOLTAGE. This procedure must be performed by a qualified electrical technician due to the danger of electrical shock, injury or even death. See your authorized dealer.

To test the elements:

1. Disconnect elements one at a time from the connection point in the screed, keeping track of connector placement.
2. Using an ohm meter, test the resistance through the element between every combination of ins (i.e., 1 to 2, 2 to 3, 1 to 3) in the plug. **(Figure 5-26)**

NOTE: You do not have to test the plug attached to the lower side of the control box.

3. Before replugging the element, check each wire pin with an ohm meter test lead, placing the lead on a steel section of the screed frame. If there is any continuity through the element to the frame, the element must be replaced.

Table 5-2. LB5 Screed Element Resistance Table

LB5 SCREED (Standard) 220 VAC	
Generator Size (kW)	14.4 kW
Breaker Size	60 Amp
Main Screed Elements (2)	
Wattage Per Element	3500 W
Total Wattage	7000 W
Resistance per Element	15 - 16.8 Ω
Screed Extension Elements (2)	
Wattage Per Element	2000 W
Total Wattage	4000 W
Resistance per Element	26 - 32 Ω
Heated End Gate Elements (2) **	
Wattage Per Element	500
Total Wattage	1000
Resistance per Element	88 - 108 Ω

**** Option item.**

Table 5-3. LB7 Screed Element Resistance Table

LB5 SCREED (Option) 480 VAC	
Generator Size (kW)	34 kW (3-Phase)
Breaker Size	63 Amp
Main Screed Elements (4)	
Wattage Per Element	3500 W
Total Wattage	14,000 W
Resistance per Element	94 - 101 Ω
Screed Extension Elements (2)	
Wattage Per Element	3000 W
Total Wattage	6000 W
Resistance per Element	108 - 118 Ω
Heated End Gate Elements (2) **	
Wattage Per Element	1000
Total Wattage	2000
Resistance per Element	325 - 335 Ω

**** Option item.**

Generator Speed Check

NOTICE **The generator should never be operated at a speed of 2100 RPMs or greater.**

A voltmeter set to read frequency (Hz) can be used to test generator speed. When the engine is running at 2000 RPMs, the generator should operate at 60 Hz. Make certain that the paver engine is running at 2000 RPMs and the hydraulic temperature is at normal operating levels.

Generator Voltage Testing

The generator is belt-driven (from the engine). While the generators are different, the procedure is the same for the standard LB5 or optional LB7 screeds. **(Figure 5-23 or 5-24 on *Page 5-28*)**

DANGER **HIGH VOLTAGE. This procedure must be performed by a qualified electrical technician due to the danger of electrical shock, injury or even death. See your authorized dealer.**

Test the generator voltage at the electric heat control box:

1. Use a volt meter to measure between two of the main input wires L-1 (black wire), L-2 (white wire) or L-3 (red wire) at the terminal block, which should measure 480 ± 5 VAC. If you measure from L-1 to the frame of the paver, or ground, the voltage should be approximately 277 VAC.
2. If the voltage is lower, make certain the generator is turning the correct speed by testing with a Hz meter (valve should be 60 - 63 Hz.
3. If the voltage is still low, contact your authorized LeeBoy dealer for generator speed tuning (see next section).

5

Generator Capacitor Replacement

The two generator capacitors control and regulate voltage while the machine operates. **(Figures 5-27 and 5-28)** If the capacitor fails, the voltage will drop below normal or no output at all. It is located under the generator cover.

If experiencing a drop in voltage, replacing the capacitor will help determine if the capacitor is at fault.

To perform this procedure:

1. Shut down the machine.
2. Detach the wires on top of the capacitor.
3. Remove the capacitor and replace with new one.
4. Reattach wires.

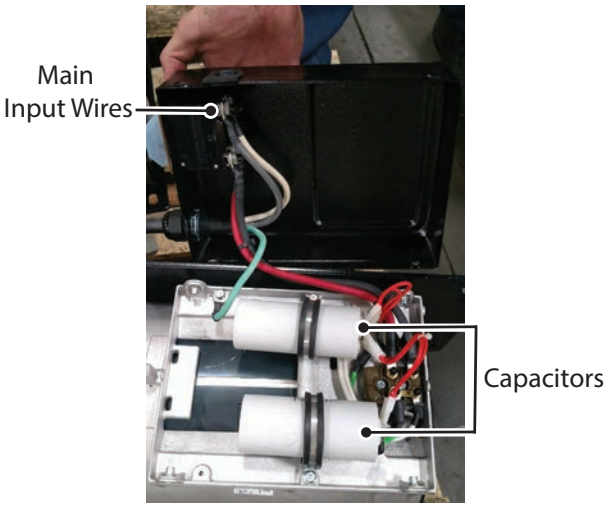


Figure 5-27. LB5 Screed Generator Capacitor

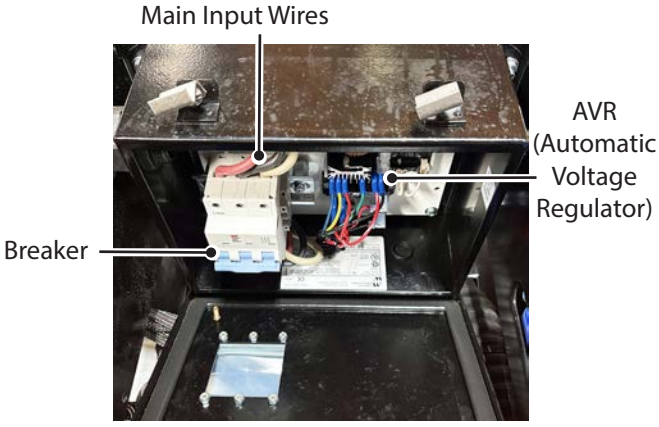


Figure 5-28. LB7 Screed Generator Capacitors

Dielectric Grease

Dielectric grease is a nonconductive, insulating compound applied to electrical connectors. When applied correctly, a high-quality dielectric grease displaces debris and moisture in the connector. The proper application of dielectric grease fills any gaps between pins or in the connector which prevents debris and moisture from creating corrosion of the electrical components.

⚠CAUTION Equipment damage can occur if using low-quality, incompatible dielectric grease.

If not applied or incorrectly applied, corrosion can result that causes rapid increase in resistance, intermittent connections, overheating, and a variety of electrical problems. Numerous electrical issues throughout the machine can be avoided by the proper application and use of dielectric grease.



Figure 5-29. Correct Application of Dielectric Grease

To correctly apply dielectric grease:

- Apply a thin, visible, uniform coat using **ONLY** high-quality dielectric grease (Dow Corning 4 or equivalent) across the **MALE** connector as shown in **Figure 5-29.**
    - Coating too thin may not fill the gaps between pins.
    - Coating too thick may prevent proper connection or force pins or seals out. **(Figure 5-30)**
- Plug in the male connector.

⚠CAUTION DO NOT force connector to lock if too much grease has been applied.

⚠CAUTION Excessive grease application does not allow a proper connection by preventing the plugs from locking into place, and can cause damage to the connector seals.



Figure 5-30. Incorrect Application of Dielectric Grease

DRIVE SYSTEM

The complete drive unit consists of the two-speed motors, sprockets, planetary torque hubs, and high-speed rubber (or the optional poly steel) track systems.

It is important to keep the undercarriage clean and free of asphalt buildup/debris. Undercarriage components should be inspected and maintained periodically, with any worn or faulty component immediately replaced or repaired for operational safety and performance. Maintenance and replacement procedures are described on the following pages.

⚠CAUTION Ensure the paver cannot be started inadvertently by others by blocking the work area before conducting maintenance or replacement procedures on undercarriage components. Use hoisting gear for lifting heavy components, and always wear protective gear.

⚠DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

For performance and durability:

- Check undercarriage components (i.e. drive wheel, rollers and idler) for wear daily. Wear and damage of undercarriage components adversely affects performance and durability.
- Limit driving on sharp, rocky surfaces, manholes and drainage covers.
- Prevent large foreign objects from getting entangled in your undercarriage.
- Remove asphalt buildup on the undercarriage components and tracks daily. Asphalt buildup can cause misalignment between the drive wheel and guide lugs, poor tracking performance, and premature undercarriage and wear.
- Avoid de-tracking episodes by:
    - Moving to a cold area to make turns.
    - Avoiding sharp turns on hot asphalt
    - Limiting movement in reverse.
    - Making gradual turns in a large arc as possible.
    - Repeating zig-zag motions with gradual turns instead of pivoting.
    - Maneuvering at low speed.

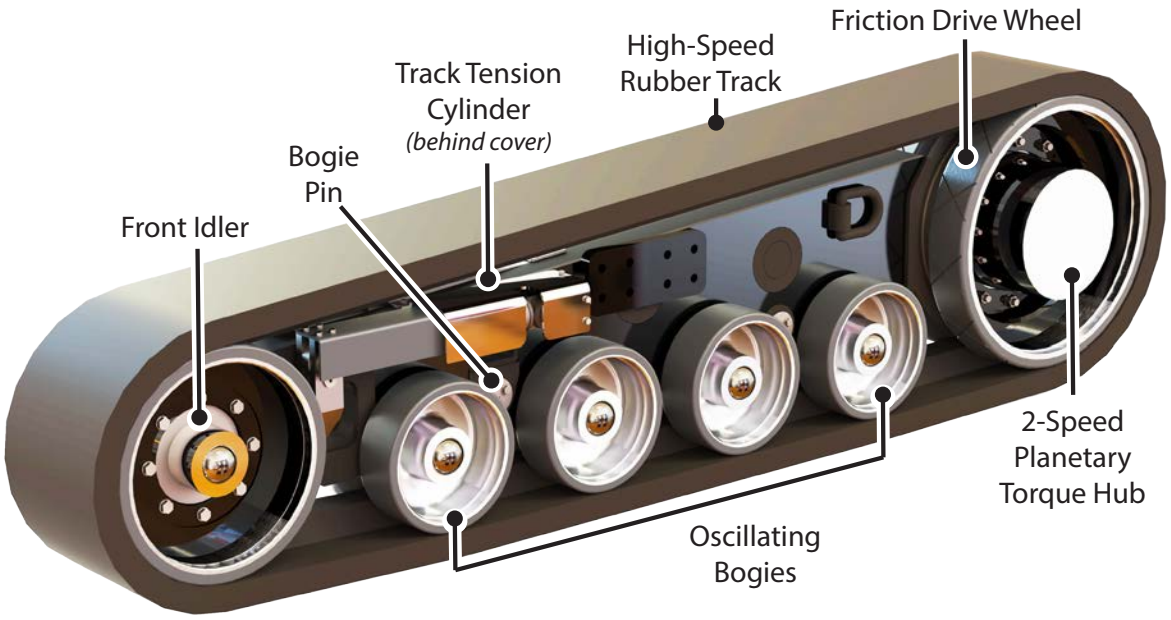


Figure 5-31. Heavy-Duty Rubber Track Components

High-Speed Rubber Tracks

Tracks may need to be replaced due to wear or damage in time, usually caused by reckless driving or load contact between undercarriage parts. Other factors that increase track wear are:

- Tracks resting on uneven rock beds during operation.
- One-sided undercarriage load during turning.
- Driving long distances in reverse.
- Driving over obstructions.
- Improper tension according to operating conditions.
- Excessive dirt and/or debris on the tracks and undercarriage.
- If equipped with the standard rubber tracks, DO NOT drive the paver with the rubber sidewalls pressing against walls, curbs or other objects.

NOTICE **If storing your paver for long periods, keep it indoors in a cool, dry environment away from rain and direct sunlight. If it must be stored outdoors, cover the tracks to reduce exposure. Operate the paver every two weeks for a few minutes to help tracks maintain flexibility and proper shape.**

Rubber track systems require little maintenance, but checking track alignment is key to ensuring longevity. It is essential that you maintain the correct tension with proper track alignment at all times.

Rubber track components are shown at **Figure 5-31.**

DANGER **Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.**

NOTICE **ALWAYS replace BOTH tracks at the same time. Replacing only one may cause alignment issues and damage the undercarriage.**

To replace the high-speed rubber tracks:

1. Park paver on a level surface.
2. Jack up the paver at least two feet off the ground for enough clearance to make it easier to lift the track into place.
3. Find the track tension manifold located inside the left engine access panel on the engine compartment floor. **(Figure 5-32)**

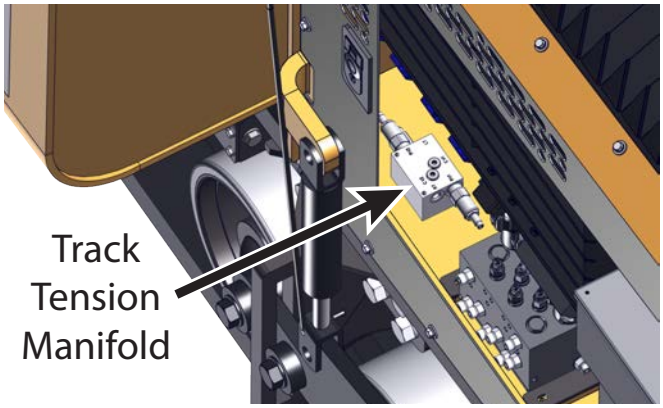


Figure 5-32. Track Tension Manifold Location

4. Loosen the RV1 (left track) or RV2 (right track) relief valve cartridge on the track tension manifold to relieve pressure by turning the cartridge 3 turns counterclockwise. **(Figure 5-33)**

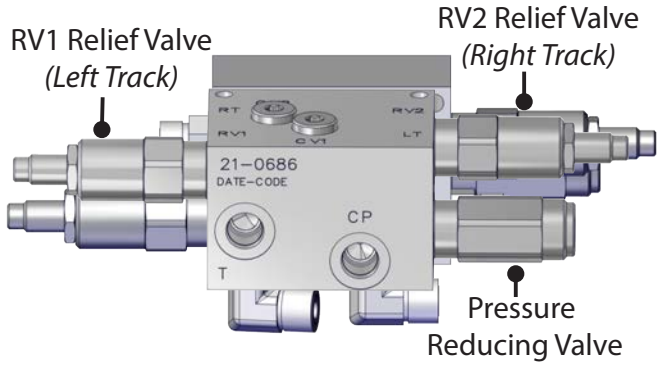


Figure 5-33. Track Tension Main Manifold

5. Remove the outer friction drive wheel (12 nuts) and outer front idler (8 bolts/washers). **(Figure 5-34)**

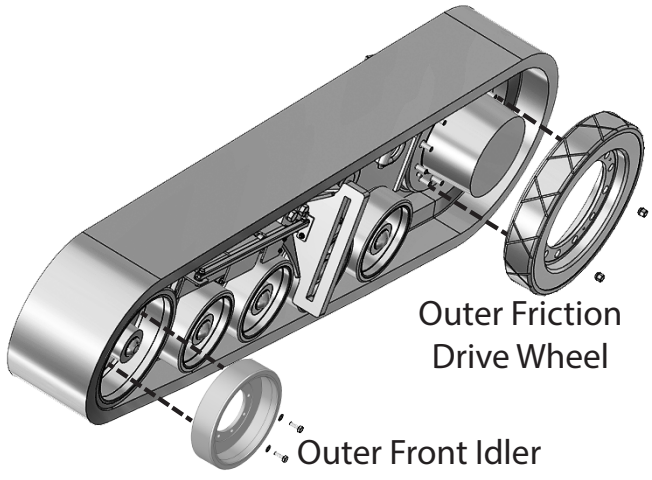


Figure 5-34. Outer Friction Drive Wheel and Outer Front Idler Removal

6. Remove rubber track.
7. Lift the new rubber track into place.
8. Reinstall the outer friction wheel and outer front idler.
9. Screw the track tension manifold relief cartridge back into place and check hydraulic oil level. Add more if needed. **(Figure 5-34)**
10. Run the machine, then re-check the tension after several hours of operation.

Track Tension Cylinder

⚠ DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

1. Follow **Steps 1 - 4** in the previous section. (The track tension cylinders are located behind the front idlers.) **(Figure 5-32)**
2. Remove cylinder covers to access cylinders and disconnect the hydraulic hoses. **(Figure 5-35 *on the following page.)***
3. Remove clevis pins from the cylinders. **(Figure 5-35)**
4. Repack the seal kit or replace cylinder if worn, and reinstall.
5. Reinstall cylinders, pins, hoses and covers.

Oscillating Bogies

⚠ DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

1. Follow the procedure for removing the track on the previous page.
2. Remove the bogie pin. **(Figure 5-31*)***
3. Tap the bogie out of the track frame and replace with new one.
4. Reinstall the bogie pin and track.